CartoGraph-X | Detailed Residual GraphSAGE and Country-Conditioned Road Selection

Exact computational operators, feature dimensions, message passing, residual pathways, gated fusion, and logit formulation

(a) Road-node input and projection

Source road segments; induced split-specific proximity graph

Node features
$X \in \mathbb{R}^{N \times F}$
23 input features

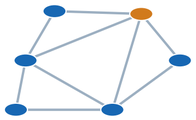
Proximity edges
$E \subseteq V \times V$
k = 6; radius = 250 m

Train-fitted preprocessing
Median imputation → StandardScaler
No test-fit preprocessing

Input projection
$z = \mathrm{ReLU}(\mathrm{LN}(W_p X + b_p))$
Linear(F, 64) → LayerNorm(64) → ReLU

(b) Exact GraphSAGE message operator

Both layers call Script 111's Layer(..., 'GraphSAGE')



Directed edge list
(src, dst)
index_add + bincount

Neighbor mean
$m_i = \sum_{j:(j,i)\in E} h_j / \max(1, d_i)$

Self + neighbor concatenation
$q_i = [h_i \,\|\, m_i] \in \mathbb{R}^{128}$

Learned combine
$\mathrm{SAGE}(h, E)_i = W_c q_i + b_c$

(c) Two residual message-passing layers

64 hidden channels; residual additions and LayerNorm

Input
$z \in \mathbb{R}^{N \times 64}$

Layer 1
$h_1 = \mathrm{LN}_1(z + D(\mathrm{ReLU}(\mathrm{SAGE}_1(z, E))))$

Layer 2
$h_2 = \mathrm{LN}_2(h_1 + D(\mathrm{ReLU}(\mathrm{SAGE}_2(h_1, E))))$

D = dropout(p = 0.20)
(training only)

Graph representation
$h = h_2 \in \mathbb{R}^{N \times 64}$

(d) Dimension-wise gated fusion of projected and contextual representations

One gate per hidden channel; graph encoder and original-feature projection are fused, not separate input encoders

Projected features
$z \in \mathbb{R}^{N \times 64}$
from projection (a)

Graph context
$h \in \mathbb{R}^{N \times 64}$
from GraphSAGE (c)

Concatenation
$u = [z \| h]$
$u \in \mathbb{R}^{N \times 128}$

Sigmoid gate
$g = \sigma(W_g u + b_g)$
$g \in (0, 1)^{N \times 64}$

Adaptive fusion
$f = g \odot h + (1 - g) \odot z$
$f \in \mathbb{R}^{N \times 64}$

$\odot$: element-wise product | σ: logistic sigmoid | g: dimension-wise mixing coefficient

(e) Shared and country-conditioned prediction

Joint-country model adds residual logits; source-only/zero-shot does not

Shared logit
$s_i = W_s f_i + b_s$
Linear(64 → 1)

Country embeddings (joint only)
$\delta_{c_i} \in \mathbb{R}^{64},\ \beta_{c_i} \in \mathbb{R}$
2 countries; embeddings initialized to zero

Source-only / zero-shot
$\ell_i = s_i$
No target-country head is trained

Joint-country
$\ell_i = s_i + f_i^{\top} \delta_{c_i} / \sqrt{64} + \beta_{c_i}$

Country index: Lithuania = 0 | Poland = 1

(f) Optimization, output and evaluation control

The figure depicts the frozen implementation, not a hypothetical network

Probabilities
$p_i = \sigma(\ell_i)$
Sigmoid after logits

Binary selection
$\hat{y}_i = \mathbf{1}[p_i \geq \tau]$
Validation-selected threshold

Training
AdamW · 60 epochs
10 seeds (42–51)

Country-balanced weighted BCE
$\mathcal{L} = \frac{1}{N} \sum w_i \,\mathrm{BCEWithLogits}(\ell_i, y_i)$
$w_i = \frac{n_c}{2 n_{c,1}}$ if $y_i = 1$; $w_i = \frac{n_c}{2 n_{c,0}}$ if $y_i = 0$ (TRAIN labels only)

All node states are 64-D after projection. Geometric kNN edges are proximity relations, not road-junction topology.